

This assignment should be done with Jupyter Notebook

1. Write a python function to price American option on trinomial tree with the following input parameters:
 - a. Spot price
 - b. Strike price
 - c. Time to maturity (in fraction of years)
 - d. Risk free rate
 - e. Dividend yield
 - f. Stock return volatility
 - g. Call or Put
 - h. Choice of trinomial schema: CRR or JR
 - i. Number of time steps
2. Carry out the following using your python pricer, and write a short paragraph to describe your observation in each of the exercises below.
 - a. On a CRR trinomial tree, price the following American call option **and** plot the early exercise boundary as a function of time.
 - i. $S = 100$
 - ii. $K = 80$
 - iii. $T = 1$
 - iv. $r = 0$
 - v. $d = 0.2$
 - vi. $\text{Vol} = 0.3$
 - vii. Number of time steps = 250
 - b. Using JR trinomial tree, price the following American put option
 - i. $S = 100$
 - ii. $K = 120$
 - iii. $T = 1$
 - iv. $R = 0.04$
 - v. $D = 0$
 - vi. $\text{Vol} = 0.3$
 - vii. Number of time steps = 250
 - c. Continue with the above put option
 - i. Plot option price vs changing spot price while holding other parameters constant: $S \in [50, 160]$

- ii. Plot option price vs changing time to maturity while holding other parameters constant: $T \in [0.25, 2]$
- iii. Plot option price vs changing volatility while holding other parameters constant: $vol \in [0.15, 1.5]$